Spark-Based Market Basket Analysis

Product Association Rules and Their Short-Term Stability

Group 6 | Project Proposal

1. Problem Description

Product associations discovered in historical orders may not persist in subsequent transactions. This project investigates two questions: Which product combinations exhibit strong co-purchase associations in historical orders? Do these associations persist in subsequent transactions, and which rules remain strong or weaken?

We will discover association rules from historical orders and assess their short-term stability using transactions from the following three months. The findings may inform product bundling or cross-selling suggestions. However, observed associations will not be interpreted as causal relationships or evidence of increased sales.

2. Tools

We plan to use HDFS for data storage and Parquet for cleaned transaction records and shopping baskets. Apache Spark, through PySpark, Spark SQL, and MLlib, will support data cleaning, basket construction, FP-Growth mining, and temporal evaluation. Matplotlib will visualize rule metrics and their monthly changes.

3. Data

We will use the publicly available Online Retail II dataset (Chen, 2012), downloaded from the UCI Machine Learning Repository. It contains approximately 1.07 million transaction line items from a UK-based online retailer between December 2009 and December 2011. Available fields include invoice number, product code, product description, quantity, unit price, and transaction timestamp. The retailer mainly sells giftware, and its customers include wholesalers.

Our analysis will focus on transactions from December 2010 to August 2011. Each valid invoice will form a shopping basket containing distinct product codes.

Chen, D. (2012). *Online Retail II* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D

4. Methodology and Algorithm

We will check for duplicate records and invoice inconsistencies, exclude cancellations, invalid sales records, and non-product entries, and aggregate the remaining transaction lines into shopping baskets.

Transactions from December 2010 to May 2011 will form the discovery period. We will apply Spark's FP-Growth algorithm to identify frequent itemsets, focusing on association rules between individual products. Rules will be selected using support, confidence, lift, and occurrence counts. Lift accounts for the consequent product's overall purchase frequency, helping distinguish associations from effects attributable to product popularity. Parameters and selection criteria will be determined using only the discovery-period data.

The selected rules will then be fixed and evaluated on transactions from June to August 2011. We will recompute the same metrics for the full evaluation period and each month, identifying associations that persist or weaken. Cases with insufficient observations, products no longer observed, or undefined metrics will be reported separately. Seasonal changes will be considered when interpreting the results.